EXPLICIT BOUNDS AND EXHAUSTIVE SEARCHES FOR FOUR-PRIME WILLIAMS NUMBERS (CARMICHAEL NUMBERS THAT ARE ALSO LUCAS–CARMICHAEL)

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ABSTRACT. A squarefree composite n is *Carmichael* if $(p-1) \mid (n-1)$ for every prime $p \mid n$, and *Lucas–Carmichael* if $(p+1) \mid (n+1)$ for every $p \mid n$. Numbers that are both—*Williams numbers*, after Williams's 1977 question—are not known to exist; they are the natural candidates for a Baillie–PSW counterexample (Pomerance) and for the Lenstra–Pomerance route to a counterexample to Agrawal's conjecture. Williams proved that such a number cannot have three prime factors, and McIntosh and Dipra (2015) proved that, with all but three prime factors fixed, only finitely many Williams numbers exist, and that a four-prime example must have smallest prime above 2×10^4 and exceed 10^{24}. For four prime factors $p_1 < p_2 < p_3 < p_4$ we give an explicit form of the finiteness theorem—all $p_i \equiv 1 \pmod{12}$, $p_3 < 2p_1p_2$, $p_4 < \sqrt{2p_1p_2p_3}$ and $p_2^2 \le 24p_1^3 + 1$, so $n < 477\, p_1^{15/2}$—and, by a complete enumeration driven by these inequalities, extend the exclusion to $n \le 10^{30}$ and to all n with $p_1 \le 500{,}000$. We also verify, by an independent factor-and-test pass over the complete tabulations, that no Carmichael number below 10^{24} is Lucas–Carmichael (308,279,939 numbers; at most four of the primes of any of them satisfy $(p+1) \mid (n+1)$). In appendices we show that the two classical construction paradigms for Carmichael numbers fail in their simplest two-sided forms (no Chernick-type polynomial family of three factors, of any number of factors with uniform offsets ± 1, or in generic position can produce such numbers; the two-sided subset-product method is measured to lose about 1.5 bits per prime), and we record heuristic estimates.

1. INTRODUCTION

Throughout, n denotes an odd squarefree composite integer with prime factorisation $n = p_1p_2\cdots p_k$, $p_1 < \cdots < p_k$, $k = \omega(n) \ge 2$.

Definition 1.1. n is a *Carmichael number* if $(p-1) \mid (n-1)$ for every $p \mid n$ (Korselt's criterion), and a *Lucas–Carmichael number* if $(p+1) \mid (n+1)$ for every $p \mid n$. A number that is both is a *Williams number* (a 1-Williams number in Echi's terminology [9]).

Carmichael numbers are the absolute Fermat pseudoprimes; Lucas–Carmichael numbers play the same role for Lucas sequences with discriminant a quadratic non-residue of every $p \mid n$. H. C. Williams [27] asked in 1977 whether a number can be both, proved that it cannot have exactly three prime factors and cannot be divisible by 3, and showed that under his Lucas-sequence formulation the number of prime factors must be odd (in the bare divisibility formulation above, all p_i lie in one class modulo 4, and k is odd when that class is 3; see Lemma 2.2). Echi [9] named such numbers *Williams numbers* (more precisely 1-Williams numbers, within a family of a-Williams numbers) and dropped the parity restriction; Bouallègue, Echi and Pinch [4] and Al-Rasasi and Ghanmi [1] studied the general a; Pinch [20] reported that there is no Williams number up to 10^{16} with an odd number of prime factors. McIntosh [17] treated the parallel problem $(p \pm 1) \mid (n-1)$ and, with Dipra [18], the Williams numbers themselves: they proved that for a fixed set of $d-3$ prime factors only finitely many Williams numbers with d prime factors exist, organised searches for them, and showed that a Williams number with exactly four

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prime factors has smallest prime $p_1 > 2 \times 10^4$ and $n > 10^{24}$. Zhang [26] developed counting heuristics for Williams numbers built from a fixed set of primes, exhibited a set of 1,029 primes expected to contain many of them, and offered a prize for the first example. No Williams number is known.

The question is tied to primality testing. Pomerance's 1984 heuristic argument that counterexamples to the Baillie–PSW primality test exist [22], restated in [5], builds $n = \prod p$ from primes with $p \equiv 3 \pmod 8$, $(5/p) = -1$ and smooth squarefree $(p-1)/2$, $(p+1)/4$, chosen so that $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ for all $p \mid n$; such an n is a Williams number, a connection made explicit in [17, 18, 26]. The construction attempted by Chen and Greene [6] for Grantham's \$620 problem [13] uses the weaker order conditions $\mathrm{ord}_p(2) \mid M$; the related set of 2,030 primes they discuss does satisfy $(p-1) \mid M$, $(p+1) \mid N$, and an unrefereed report by Phatak [19] states that no subset of that set is a Williams number. The Lenstra–Pomerance analysis of Agrawal's conjecture from the AKS primality test [2, 16] shows that the natural counterexamples are Williams numbers whose primes are all $\equiv 3 \pmod{80}$. Howe's higher-order Carmichael numbers [14] are a further neighbour: by Howe's criterion, a squarefree n is Carmichael of order 2 iff for every $p \mid n$ one has $n \equiv 1$ or $n \equiv p \pmod{p^2-1}$, while n is a Williams number iff $n \equiv p \pmod{(p^2-1)/2}$ for every $p \mid n$ (Lemma 2.1). We use the name "Williams number" throughout; the abbreviation CLC (Carmichael and Lucas–Carmichael) in the code and the earlier notes means the same thing.

Our results are the following.

- **Four prime factors** (Theorems 3.1 and 3.8). If $n = p_1p_2p_3p_4$ is a Williams number then all $p_i \equiv 1 \pmod{12}$, $p_3 < 2p_1p_2$, $p_4 < \sqrt{2p_1p_2p_3}$, and $p_2^2 \le 24p_1^3+1$; in particular $n < 477\,p_1^{15/2}$. The finiteness for fixed p_1 is McIntosh–Dipra's theorem for $d = 4$, and the mechanism of the proof (three divisibility quotients whose product is pinned near $8p_1^3$) is the one McIntosh introduced in [17]; what we add is the explicit and, as far as we can tell, sharper form of the inequalities, and we use them to drive a complete enumeration: no four-prime Williams number has $n \le 10^{30}$, and none exists with $p_1 \le 500{,}000$, extending McIntosh–Dipra's $n > 10^{24}$, $p_1 > 2 \times 10^4$.
- **Verified bound** (Theorem 4.1). No Williams number of any number of prime factors is smaller than 10^{24}. This is a complete factor-and-test pass over the tabulations of all Carmichael numbers below 10^{24}, with completeness and per-class cross-checks against independently published counts. The largest number of primes $p \mid n$ with $(p+1) \mid (n+1)$ attained below 10^{24} is four (first at a number with $\omega(n) = 9$); 826 Carmichael numbers attain exactly three.
- **Heuristics** (§5, speculative). A simple probabilistic model puts the expected number of four-prime Williams numbers near 10^{-6}, concentrated at $p_1 = 13$.
- **Appendices.** Appendix A shows that the two classical constructions of Carmichael numbers fail in their simplest two-sided forms: Chernick-type polynomial families cannot produce Williams numbers with three factors, with uniform offsets ± 1, or in generic position (Theorems A.1–A.3), the degenerate strata are empty in the ranges searched, and the two-sided subset-product method is measured to lose about 1.5 bits per prime. Appendix B gives a heuristic onset estimate.

All code, logs and result files are available (§6), together with a certification record for the enumerations.

2. Preliminaries

For an odd prime p write $L_p = (p^2-1)/2 = \mathrm{lcm}(p-1, p+1)$.

Lemma 2.1. *Let $p \mid n$. Then $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ hold together if and only if $n/p \equiv 1 \pmod{L_p}$, equivalently $n \equiv p \pmod{L_p}$.*

Proof. The two divisibilities say $n \equiv 1 \pmod{p-1}$ and $n \equiv -1 \pmod{p+1}$. Since $p \equiv 1 \pmod{p-1}$ and $p \equiv -1 \pmod{p+1}$, this is $n \equiv p$ modulo $\operatorname{lcm}(p-1, p+1) = L_p$ (the gcd of $p \mp 1$ is 2). Finally $L_p \mid n - p = p\,(n/p - 1)$ with $\gcd(p, L_p) = 1$ gives $L_p \mid n/p - 1$, and conversely. □

Lemma 2.2 (congruence classes)**.** *Let n be a Williams number with k prime factors.*

(a) *All $p \mid n$ lie in one class c modulo 4, and $n \equiv c \pmod 4$; hence k is odd if $c = 3$.*
(b) *$3 \nmid n$; all $p \mid n$ lie in one class c' modulo 3, and $n \equiv c' \pmod 3$; hence k is odd if $c' = 2$.*
(c) *If k is even, every $p \mid n$ is $\equiv 1 \pmod{12}$.*

Proof. (a) $8 \mid p^2 - 1$ for odd p, so $4 \mid L_p$ and $n \equiv p \pmod 4$ for every $p \mid n$ by Lemma 2.1. Thus all $p \equiv c$, and $n \equiv c^k \pmod 4$; if $c = 3$ this forces k odd. (b) For $p \neq 3$, $3 \mid p^2 - 1$, so $3 \mid L_p$ and $n \equiv p \pmod 3$. If $3 \mid n$ then every other prime factor would be $\equiv n \equiv 0 \pmod 3$, impossible for $k \geq 2$; so $3 \nmid n$, all $p \equiv c' \pmod 3$ and $n \equiv c'^k \equiv c'$, which for $c' = 2$ forces k odd. (c) is (a) and (b) with k even. □

Williams proved that $k = 3$ is impossible. For completeness we give a four-line proof in the present notation.

Proposition 2.3 (Williams)**.** *No Williams number has exactly three prime factors.*

Proof. Let $n = p_1p_2p_3$. By Lemma 2.1 at p_3, $p_1p_2 - 1 = tL_3$ with $t \geq 1$ (as $p_1p_2 > 1$); since $p_1p_2 < p_3^2$ we have $t < 2$, so $2p_1p_2 = p_3^2 + 1$. At p_2, $p_1p_3 - 1 = sL_2$ with $s \geq 1$; now $p_3 < \sqrt{2p_1p_2} < \sqrt{2}\,p_2$, so $p_1p_3 < \sqrt{2}\,p_2^2$ and $s < 2\sqrt{2}\,p_2^2/(p_2^2 - 1) < 3$ (as $p_2 \geq 5$). If $s = 2$ then $p_1p_3 = p_2^2$, impossible; so $2p_1p_3 = p_2^2 + 1$. Subtracting, $2p_1(p_2 - p_3) = p_3^2 - p_2^2$, i.e. $2p_1 = -(p_2 + p_3) < 0$, a contradiction. □

3. Four prime factors

3.1. **Structure.**

Theorem 3.1. *Let $n = p_1p_2p_3p_4$, $p_1 < p_2 < p_3 < p_4$, be a Williams number, and write $P = p_1p_2$, $q = p_3$, $r = p_4$.*

(a) *Every $p_i \equiv 1 \pmod{12}$; in particular $p_1 \geq 13$ and $p_2 \geq 37$.*
(b) *There are integers $d, e, s \geq 1$ with*

$$2Pq = d(r^2 - 1) + 2, \qquad 2Pr = e(q^2 - 1) + 2, \qquad 2p_1qr = s(p_2^2 - 1) + 2, \tag{1}$$

and consequently $q < 2P$, $r < \sqrt{2Pq} < 2P$, $m := qr < 4P^2$, $n < 4P^3$ and $d < 2p_1$.
(c) *$p_2^2 \leq 24p_1^3 + 1$.*

Consequently $p_2 < 4.9\,p_1^{3/2}$, $p_3, p_4 < 2P < 9.8\,p_1^{5/2}$ and $n < 477\,p_1^{15/2}$.

Proof. (a) is Lemma 2.2(c).

(b) Lemma 2.1 at r gives $n/r - 1 = Pq - 1 = dL_r$ with an integer $d \geq 1$, which is the first equation; the second and third are Lemma 2.1 at q and at p_2. From $2Pq = d(r^2-1)+2 \geq r^2+1 > q^2$ we get $q < 2P$ and $r^2 < 2Pq$, hence $r < \sqrt{2Pq} < 2P$, $m = qr < 4P^2$ and $n = Pm < 4P^3$. Also $d = 2(Pq-1)/(r^2-1) < 2Pq/(r^2-1) < 2P/q < 2p_1$ because $r \geq q+1$ and $q > p_2$.

(c) Multiply the three equations in (1). The left side is $8P^2p_1\,q^2r^2 = 8p_1^3p_2^2q^2r^2$. Writing $d(r^2-1)+2 = dr^2\big(1 - \frac{d-2}{dr^2}\big)$ and similarly for the other two factors, the right side is $d\,e\,s\;r^2q^2p_2^2\,\Pi$ with

$$\Pi = \Big(1 - \frac{d-2}{dr^2}\Big)\Big(1 - \frac{e-2}{eq^2}\Big)\Big(1 - \frac{s-2}{sp_2^2}\Big).$$

Hence

$$d\,e\,s\cdot\Pi = 8p_1^3. \tag{2}$$

For an integer $t \geq 1$ and real $x > 0$ one has $-1/x^2 \leq (t-2)/(tx^2) < 1/x^2$, so each factor of Π lies in $[1-1/x^2,\, 1+1/x^2]$ with $x \in \{r, q, p_2\}$, $x \geq p_2$; therefore

$$(1-p_2^{-2})^3 \leq \Pi \leq (1+p_2^{-2})^3. \tag{3}$$

Now reduce the three equations (1) modulo p_1. Each left side is divisible by p_1, so $d(r^2-1) \equiv e(q^2-1) \equiv s(p_2^2-1) \equiv -2 \pmod{p_1}$. As p_1 is odd, $p_1 \nmid d$, $p_1 \nmid e$ and $p_1 \nmid s$. Hence $des \neq 8p_1^3$, i.e. $\Pi \neq 1$.

Case $\Pi < 1$. Then $des \geq 8p_1^3+1$ and $\Pi = 8p_1^3/des \leq 1-\frac{1}{8p_1^3+1}$; with $\Pi \geq (1-p_2^{-2})^3 \geq 1-3p_2^{-2}$ this gives $p_2^2 \leq 3(8p_1^3+1) = 24p_1^3+3$. Since $p_2 \equiv 1 \pmod{12}$ implies $p_2^2 \equiv 1 \pmod{24}$, while $24p_1^3+2$ and $24p_1^3+3$ are $\equiv 2, 3 \pmod{24}$, in fact $p_2^2 \leq 24p_1^3+1$.

Case $\Pi > 1$. A factor $1-\frac{t-2}{tx^2}$ of Π exceeds 1 only when $t=1$. Now $e > d \geq 1$, because $e/d = \frac{(Pr-1)(r^2-1)}{(Pq-1)(q^2-1)} > 1$ for $r > q$; and $s > 2p_1$, because $qr \geq p_2^2+1$ gives $s = \frac{2(p_1qr-1)}{p_2^2-1} \geq \frac{2(p_1p_2^2+p_1-1)}{p_2^2-1} > 2p_1$. So only the first factor can exceed 1, and $\Pi \leq 1+r^{-2}$. On the other hand $des \leq 8p_1^3-1$ gives $\Pi = 8p_1^3/des \geq 1+\frac{1}{8p_1^3-1}$. Hence $r^2 \leq 8p_1^3-1$, and $p_2^2 < r^2 < 8p_1^3 \leq 24p_1^3+1$.

In both cases $p_2^2 \leq 24p_1^3+1$.

The consequences follow from (b) and (c): $p_2^2 \leq 24p_1^3+1 < 24.01\,p_1^3 = (4.9\,p_1^{3/2})^2$ for $p_1 \geq 5$, so $p_2 < 4.9\,p_1^{3/2}$; then $q, r < 2P < 9.8\,p_1^{5/2}$ and $n < 4P^3 < 4\cdot 4.9^3\,p_1^{15/2} < 477\,p_1^{15/2}$. □

Remark 3.2. An earlier draft handled the case $\Pi > 1$ with the estimate $(1+p_2^{-2})^3 \leq 1+3.0001\,p_2^{-2}$, which is false (the correct coefficient at $p_2 = 37$ is 3.0022, and with it that route does not close); the error was found in the independent check described in §6. The enumeration of §3.2 used the bound $p_2^2 \leq 24p_1^3+3$, which both cases satisfy, so Theorem 3.8 is unaffected.

Corollary 3.3. *For each prime p_1 there are only finitely many four-prime Williams numbers with smallest prime p_1, all below $477\,p_1^{15/2}$. Conversely a four-prime Williams number n has $p_1 > (n/477)^{2/15}$.*

Remark 3.4 (relation to McIntosh and Dipra). The finiteness statement of Corollary 3.3 is the case $d=4$ of McIntosh and Dipra's theorem [18], and the argument—three quotient equations whose product is squeezed around $8p_1^3$ while the product of the quotients cannot equal $8p_1^3$—is the one McIntosh used for the parallel problem $(p\pm1) \mid (n-1)$ in [17], where it yields a bound of the shape $q < 20P^{5/2}$ for the second free prime. The contribution of Theorem 3.1 is the explicit form: the exponent $3/2$ in $p_2 < 4.9\,p_1^{3/2}$ and the constants, which make the enumeration of §3.2 feasible far beyond 10^{24}. **[TODO: compare with the explicit inequalities and constants stated in [18] once the paper has been read in full; adjust "sharper" accordingly.]**

3.2. **Enumeration.** Theorem 3.1 makes two kinds of complete enumeration possible: all $n \leq B$ (where $n > p_1p_2^3$ forces $p_1 < B^{1/4}$ and $p_2 < (B/p_1)^{1/3}$), and all n whatsoever with $p_1 \leq X$ (where $p_2 \leq \sqrt{24p_1^3+1}$). Both reduce to a search over pairs (p_1, p_2).

Lemma 3.5 (one class for the cofactor)**.** *With notation as in Theorem 3.1, $m = qr$ satisfies $p_2m \equiv 1 \pmod{L_{p_1}}$ and $p_1m \equiv 1 \pmod{L_{p_2}}$. These are solvable iff $p_1 \nmid p_2^2-1$ and $p_1 \equiv p_2 \pmod{\gcd(L_{p_1}, L_{p_2})}$, in which case m lies in a single residue class modulo $\Lambda = \operatorname{lcm}(L_{p_1}, L_{p_2})$.*

Proof. The congruences are Lemma 2.1 at p_1 and p_2 ($n/p_1 = p_2m$, $n/p_2 = p_1m$). The first congruence is always solvable: $p_2 \mid L_{p_1}$ would need $p_2 \mid p_1 \pm 1$ with $p_2 > p_1$, impossible for

odd primes. The second is solvable iff $\gcd(p_1, L_{p_2}) = 1$, i.e. $p_1 \nmid p_2^2 - 1$. Given both, let $g = \gcd(L_{p_1}, L_{p_2})$; the pair is compatible iff $p_2^{-1} \equiv p_1^{-1} \pmod g$, i.e. $p_1 \equiv p_2 \pmod g$ (note $\gcd(p_1 p_2, g) = 1$). The rest is the Chinese remainder theorem. □

Since $\Lambda \approx P^2/(4g)$ while $m < 4P^2$, Lemma 3.5 leaves $O(g)$ candidates m per pair. The next lemma rejects almost all of them with one multiplication.

Lemma 3.6 (window)**.** *With d, e as in (1) and $j := de$, $T := 4P^2$,*

$$|jm - T| < \max\bigl(2j\sqrt{2p_1},\ 8p_1\bigr). \tag{4}$$

Moreover, for fixed $m > 2\lceil\sqrt{2p_1}\rceil$ (which holds for $m > p_2^2$) the set of real $j > 0$ satisfying $|jm - T| \le \max(2j\lceil\sqrt{2p_1}\rceil, 8p_1)$ is an interval containing T/m; hence it contains an integer iff it contains $\lfloor T/m\rfloor$ or $\lceil T/m\rceil$.

Proof. From (1), $d = 2(Pq-1)/(r^2-1)$ and $e = 2(Pr-1)/(q^2-1)$, so $j\,(q^2-1)(r^2-1) = 4(Pq-1)(Pr-1)$. Expanding with $m = qr$,

$$jm^2 - Tm = j(q^2 + r^2 - 1) - 4P(q+r) + 4.$$

The right side is a difference of two positive numbers, so its absolute value is at most $\max\bigl(j(q^2+r^2-1),\ 4P(q+r)-4\bigr) \le \max(2jr^2,\ 8Pr)$. Dividing by $m = qr$ gives $|jm - T| \le \max(2jr/q,\ 8P/q)$. By Theorem 3.1(b), $r^2 < 2Pq$, so $r/q < \sqrt{2P/q} < \sqrt{2p_1}$ (as $q > p_2$), and $P/q < p_1$; this is (4). For the interval property, put $S = \lceil\sqrt{2p_1}\rceil$ and $\phi(j) = |jm - T| - \max(2jS, 8p_1)$. On $j \ge T/m$ the first term increases with slope m and the second with slope at most $2S < m$, so ϕ is strictly increasing; on $j \le T/m$ the first term decreases and the second is non-decreasing, so ϕ is strictly decreasing. Thus $\{\phi \le 0\}$ is an interval, and it contains T/m since $\phi(T/m) < 0$. □

Algorithm. Put $S = \lceil\sqrt{2p_1}\rceil$, $R_j = \max(2jS, 8p_1)$, and call (m, j) *admissible* if $|jm - T| \le R_j$; by Lemma 3.6 the true (m, de) is admissible, and for fixed $m > p_2^2 > 2S$ the admissible j form an interval containing T/m. For every pair of primes $p_1 < p_2$, both $\equiv 1 \pmod{12}$, within the range dictated by the campaign: (i) skip the pair unless both conditions of Lemma 3.5 hold (a pair failing them has no CLC completion), and compute m_0, Λ; (ii) the candidates are $m \equiv m_0 \pmod\Lambda$ with $p_2^2 < m \le m_{\max} := \min(4P^2 - 1, \lfloor B/P\rfloor)$; choose a threshold $M^* \approx \sqrt{T\Lambda}$; for $m \le M^*$ walk the candidates and keep m iff $\lfloor T/m\rfloor$ or $\lceil T/m\rceil$ is admissible for it; for $m > M^*$ walk instead the integers j from $\max\bigl(1, \lfloor(T-8p_1)/(m_{\max}+2S)\rfloor - 1\bigr)$ to $\lfloor(T+8p_1)/(M^*-2S)\rfloor + 1$ and keep every candidate $m \in (M^*, m_{\max}]$ in the exact window $\lceil(T-R_j)/j\rceil \le m \le \lfloor(T+R_j)/j\rfloor$ (any admissible (m, j) with $m > M^*$ has j in that range, since $jm \le T + 2jS + 8p_1$ and $jm \ge T - 2jS - 8p_1$); the two walks were checked to produce identical survivor lists on six instances, including two relaxed instances with 346,409 and 11,007,875 survivors; (iii) for each surviving m require $2^{n-1} \equiv 1 \pmod n$, $n = Pm$ (necessary for a Carmichael number; computed in C with 128-bit modular arithmetic when $P \le 2^{40}$, and otherwise deferred to the Python verifier below); (iv) for each remaining m, recover r as an integer root of $dx^3 - (d-2)x - 2n$ for $d = 1, \dots, 2p_1$ (multiply the first equation of (1) by r: $2n = dr^3 - (d-2)r$; Theorem 3.1(b) gives $d < 2p_1$), set $q = m/r$, and check all four conditions of Lemma 2.1 together with the primality of q and r. Independently of the C program, a Python program using exact integer arithmetic and SymPy factorisation re-tests every survivor of step (ii) that it is given.

Proposition 3.7 (completeness of the search)**.** *Let $n = p_1 p_2 q r$ be a Williams number with four prime factors, and suppose $n \le B$ (campaign 1) or $p_1 \le X$ (campaign 2). Then the algorithm above visits the pair (p_1, p_2), the candidate $m = qr$ survives step (ii) and step (iii), and step (iv) recovers (q, r) and reports n.*

Proof. By Theorem 3.1(a) both p_1, p_2 are primes $\equiv 1 \pmod{12}$. If $n \le B$ then $p_1^4 < p_1 p_2 q r = n \le B$ and $p_1 p_2^3 < n \le B$, so $p_1 < B^{1/4}$ and $p_2 < (B/p_1)^{1/3}$; in either campaign $p_2 \le \sqrt{24 p_1^3 + 1}$ by Theorem 3.1(c). Hence the pair is in the range scanned. The congruences of Lemma 3.5 hold for m, so the pair passes step (i) and $m \equiv m_0 \pmod{\Lambda}$; by Theorem 3.1(b), $p_2^2 < m < 4P^2$, and $m \le B/P$ when $n \le B$, so m is one of the candidates walked in step (ii). By Lemma 3.6 the pair (m, de) is admissible, so if $m \le M^*$ then $\lfloor T/m \rfloor$ or $\lceil T/m \rceil$ is admissible and m is kept, while if $m > M^*$ then $j = de$ lies in the scanned j-range and m lies in the exact window for that j, so again m is kept. Since n is a Carmichael number, $2^{n-1} \equiv 1 \pmod{n}$ and m survives step (iii) (or is deferred to the Python verifier, which applies the same test). Finally $2n = dr^3 - (d-2)r$ with $1 \le d < 2p_1$ (Theorem 3.1(b)), so the loop over d in step (iv) reaches d, the integer cube root recovers r, $q = m/r$, and all checks pass. The actual runs are certified by the record described in §6: every block of pairs logged exactly once with contiguous p_1 ranges, and every window survivor re-tested in Python. □

Validation. The implementation was tested against brute force on a *relaxed* problem that does have solutions: replace L_p by $(p^2 - 1)/\kappa$ for $\kappa \in \{24, 120\}$. Theorem 3.1(b) and Lemmas 3.5–3.6, which are all the algorithm uses, go through with 2 replaced by κ (Theorem 3.1(c) and the Fermat filter are specific to $\kappa = 2$ and are switched off in the relaxed runs, so step (iii) is validated only by the Python re-verification). On the relaxed problem the enumerator finds exactly the solution set that a direct four-fold loop finds (for $\kappa = 24$ the unique solution below 10^{12} is $13685 = 5 \cdot 7 \cdot 17 \cdot 23$, for which $d = 27$, $e = 67$, $s = 977$ and $des \cdot \Pi = 24^3 \cdot 5^3$ exactly; the outputs of both programs to 10^{12} are archived in `clc/k4/validation/`). This is a regression test of the implementation on an instance with known answer, not an independent validation of the production search, which rests on the completeness argument above and the certification record of §6. Separate self-tests brute-force the two-prime sub-problem (q, r) for small (p_1, p_2) in four relaxed variants (43 solutions) and check every inequality used above, and brute-force the residue class of Lemma 3.5 for 13,037 pairs. On the real problem the enumerator reproduces the empty result below 10^{24} of Theorem 4.1 in under ten seconds. One design error was caught by these tests before any production run: the true $j = de$ is *not* in general $\lfloor T/m \rfloor$ or $\lceil T/m \rceil$ (only the filter may use those two values), so the reconstruction step (iv) must not depend on j.

Theorem 3.8. *There is no Williams number with exactly four prime factors and $n \le 10^{30}$, and there is none at all whose smallest prime factor is at most* 500,000*. Consequently a four-prime Williams number, if one exists, has $p_1 >$ 500,000 and $n > 10^{30}$. (McIntosh and Dipra* [18] *had $p_1 > 2 \times 10^4$ and $n > 10^{24}$.)*

Computation. Table 1 summarises the two campaigns (Apple M4 Pro, 14 threads). In each, the single survivor of the Fermat filter is the Carmichael number $26921089 = 13 \cdot 37 \cdot 97 \cdot 577$, for which $38 \nmid 26921090$. Both campaigns were then rerun with every window survivor written out, and certified by a script with hard assertions (`clc/k4/certify_k4.py`): every block of pairs logged exactly once, block ranges contiguous over the admissible primes, per-block counts equal to the production run, and all 1,904 + 1,185 distinct window survivors re-tested in Python (base-2 Fermat test, then factorisation and the four divisibilities on the original tuple); exactly one survivor in each campaign passes the Fermat test, the Carmichael number above, and none is a Williams number. The certification records, with hashes of the binary, source and logs, are `CERTIFY_n1e30.txt` and `CERTIFY_p1_500000.txt`. The second statement combines the second campaign with Corollary 3.3. □

campaign	pairs (p_1, p_2)	window survivors	Fermat survivors	solutions	wall time
all $n \le 10^{30}$	230,612,582,193	1,904	1	0	87 min
all n, $p_1 \le 500{,}000$	86,517,004,140	1,185	1	0	146 min

TABLE 1. The four-prime enumeration (Theorem 3.8).

4. No Williams number below 10^{24}

Theorem 4.1. *No Williams number is smaller than* 10^{24}*. Among the* 308,279,939 *Carmichael numbers below* 10^{24}*, the largest number of prime factors* p *with* $(p+1) \mid (n+1)$ *is four, first attained at*

$$155003096905548657580 09 = 37 \cdot 103 \cdot 109 \cdot 127 \cdot 307 \cdot 523 \cdot 613 \cdot 1429 \cdot 2089$$

(the primes $37, 109, 1429, 2089$*); 826 numbers have exactly three. Below* 10^{22} *the maximum is three, attained by* 179 *numbers. (Our scan recorded the first number attaining the maximum and the count of three-match numbers; the number of four-match numbers was not recorded and would require a rerun.)*

This is, as far as we have found, the largest explicit bound in print for Williams numbers with an arbitrary number of prime factors; Pinch's 10^{16} [20] is restricted to an odd number of prime factors, and McIntosh and Dipra's 10^{24} [18] to four. **[TODO: check whether** [18] **also reports a complete search for all** d **(their companion paper** [17] **does so for the** $(p \pm 1) \mid (n-1)$ **problem up to** 10^{24}**).]**

Method. Carmichael numbers have been tabulated completely to 10^{21} by Pinch [20, 21], to 10^{22} by Goutier [12] and to 10^{24} by Shallue and Webster [23, 24], whose list is distributed by Stamm-Wilbrandt as a PARI/GP binary vector [25]. A Williams number is in particular Carmichael, so it suffices to test every tabulated Carmichael number for the Lucas–Carmichael property. We did this ourselves rather than rely on flags computed by the tabulators: each n was factored (the Miller–Rabin square-root-of-unity splitting, which always succeeds on Carmichael numbers, with a general factoring fallback), the factorisation was checked to be squarefree with product n and to satisfy Korselt's criterion, and the number of primes p with $(p+1) \mid (n+1)$ was recorded. Processing is line-based so that no entry can be skipped at a buffer boundary. Three completeness checks were applied to the 10^{24} list: the counts below 10^{22}, 10^{23} and 10^{24} (49,679,870; 123,381,982; 308,279,939) equal the published ones [24], the list is strictly increasing, and it begins $561, 1105, 1729, 2465, 2821$. Our per-$\omega$ tallies agree with independently published counts: 1,494,738 three-prime Carmichael numbers below 10^{24} (OEIS A132195, [8]) and 6,529,921 eleven-prime ones (OEIS A299711); below 10^{22} all eleven per-ω counts agree with Goutier's tally, and the 179 three-match numbers below 10^{22} agree with OEIS A329948 (Suteu; b-file by Eldar). The converse direction was also checked: none of the 129,125 Lucas–Carmichael numbers below 10^{15} tabulated by Suteu (OEIS A006972) is Carmichael. Table 2 gives the distribution below 10^{24}.

Corollary 4.2. (a) *A counterexample to Agrawal's conjecture of the Lenstra–Pomerance type (a Williams number all of whose primes are* $\equiv 3 \pmod{80}$*) exceeds* 10^{24}*.*

(b) *A Baillie–PSW counterexample of the Pomerance type (a Carmichael number with* $(p+1) \mid (n+1)$ *for all* $p \mid n$*) exceeds* 10^{24}*.*

Statement (b) adds little to the known absence of Baillie–PSW counterexamples below $2^{64} \approx 1.8 \times 10^{19}$ [11], but it is a different kind of statement: it bounds every construction that proceeds through the Pomerance conditions, not only the tested range.

$\omega(n)$	count	$\omega(n)$	count	$\omega(n)$	count
3	1,494,738	7	62,338,997	11	6,529,921
4	548,265	8	99,469,866	12	547,590
5	3,429,390	9	81,041,495	13	15,328
6	19,665,194	10	33,199,094	14	61

TABLE 2. Carmichael numbers below 10^{24} by number of prime factors, as recounted by our scan (total 308,279,939).

Remark 4.3 (Howe's order-2 Carmichael numbers). By Howe's criterion [14], a squarefree n is a Carmichael number of order 2 iff for each $p \mid n$, $n \equiv 1$ or $n \equiv p \pmod{p^2-1}$. For a Williams number the first option is impossible ($n \equiv p \pmod{L_p}$ and $n \equiv 1 \pmod{p^2-1}$ would give $L_p \mid p-1$), so a Williams number is of order 2 iff $n \equiv p \pmod{p^2-1}$ for all $p \mid n$, i.e. iff $(n-p)/L_p$ is even for every $p \mid n$. Tabulations of order-2 Carmichael numbers therefore bound one 2-adic type of Williams number; we have not pursued this.

5. Heuristics

This section is speculative: it records what a simple probabilistic model predicts, as motivation and as a consistency check against the computations. It proves nothing and is not used elsewhere in the paper.

Four prime factors. Condition on Theorem 3.1(a). For $p \equiv 1 \pmod{12}$ the condition $n/p \equiv 1 \pmod{L_p}$ is automatic modulo 12, so model the remaining condition as an event of probability $24/p^2$, independently for the four primes. Summing $\prod_i 24/p_i^2$ over quadruples of primes $\equiv 1 \pmod{12}$ with $p_2^2 \le 24p_1^3+1$ and $p_2 < q < r < 2p_1p_2$ (Theorem 3.1) gives an expected count of about 6.3×10^{-7} (script `agrawal/k4_heuristic.py`), of which about 96% comes from $p_1 = 13$ (where $n < 1.1 \times 10^{11}$, long excluded by Theorem 4.1); the contribution of $p_1 > X$ decays like X^{-4} (up to logarithms) and is of order 10^{-25} for $X = 500{,}000$. On this model the four-prime case is empty with overwhelming probability, and Theorem 3.8 confirms the part of the prediction that can be tested.

Onset of Williams numbers of any size. A cruder Erdős–Pomerance-style model for arbitrary numbers of prime factors is described in Appendix B; it suggests that the first Williams number, if any, has hundreds of digits and on the order of a hundred prime factors.

6. Reproducibility

All code is in the repository `https://github.com/razv2025/carmichael-lucas-carmichael`, tagged for this version. The 10^{24} scan (`clc/`) consists of the download–decode–verify–scan pipeline, the completeness report, the result files with per-ω and per-match counts, and SHA-256 checksums of the input data. The four-prime enumeration (`clc/k4/`) consists of the C program, its brute-force reference, the Python re-verifier, the per-block logs of both campaigns (one JSON line per block, so the coverage can be audited), the survivor dumps and the certification records `CERTIFY_*.txt`, and `make test`, which runs the self-tests and the relaxed end-to-end comparison in under a minute. The family searches and the heuristic model are in `agrawal/`, with a PYTEST suite that includes symbolic checks of the identities used in the proofs of Theorems 3.1 and A.1–A.3. The consultation history, including the errors caught along the way, is in `docs/`.

7. Open problems

(1) Is $k = 4$ impossible? By Theorem 3.1, fixing p_1 leaves a finite search over (p_2, q, r); the nonzero integer $\delta = des - 8p_1^3 = 8p_1^3(1/\Pi - 1)$ ($\delta \geq 1$ when $\Pi < 1$, $\delta \leq -1$ when $\Pi > 1$) ties the sizes of p_2, q, r to p_1 but does not by itself exclude them. McIntosh [17] observed empirically, for the parallel problem, that the product of the three quotients always exceeds $8P^3$, and that a proof would force at least six prime factors.
(2) The $k = 5$ case, where both classes modulo 4 are allowed and the analogue of (2) has four factors.
(3) Closing the $m = 4$ polynomial-family stratum with one repeated root.
(4) A rigorous two-sided analogue of Pomerance's upper bound for Carmichael numbers.

Appendix A. Obstructions to the classical constructions

Carmichael numbers are constructed in two ways: by polynomial families in the manner of Chernick [7], $n(k) = (6k+1)(12k+1)(18k+1)$, where the divisibilities are polynomial identities; and by subset products in the manner of Erdős and Alford–Granville–Pomerance [10, 3, 15], where many primes p with $p - 1$ dividing a fixed smooth Λ are harvested and a subset with product $\equiv 1 \pmod{\Lambda}$ is sought. We show that the two-sided versions of both break down in their simplest forms.

A.1. **Polynomial families.** Consider $n(k) = \prod_{i=1}^{m}(a_i k + b_i)$ with integers $a_i > 0$, the linear factors pairwise non-proportional, and require the CLC divisibilities as polynomial identities: $(a_i k + b_i - 1) \mid (n(k) - 1)$ and $(a_i k + b_i + 1) \mid (n(k) + 1)$ in $\mathbb{Q}[k]$. Since a linear polynomial divides f iff f vanishes at its root, this is $n\big((1 - b_i)/a_i\big) = 1$ and $n\big((-1 - b_i)/a_i\big) = -1$. With $D_{ij} = a_i b_j - a_j b_i$ one computes $a_j \frac{1-b_i}{a_i} + b_j = (a_j + D_{ij})/a_i$, so the conditions read, for every i,

$$\prod_{j \neq i}(a_j + D_{ij}) = a_i^{m-1} = \prod_{j \neq i}(D_{ij} - a_j). \tag{5}$$

Theorem A.1 (three factors)**.** *No such family exists for $m = 3$.*

Proof. Subtracting the two products in (5) for $i = 1, 2, 3$ gives $a_2 D_{13} + a_3 D_{12} = 0$, $a_1 D_{23} + a_3 D_{21} = 0$, $a_1 D_{32} + a_2 D_{31} = 0$, i.e. $a_2 D_{13} = -a_3 D_{12}$, $a_1 D_{23} = a_3 D_{12}$, $a_1 D_{23} = -a_2 D_{13}$. The identity $a_1 D_{23} - a_2 D_{13} + a_3 D_{12} = 0$ (it is $a \wedge b \wedge a = 0$) becomes $3a_3 D_{12} = 0$, so $D_{12} = 0$ and then $D_{13} = D_{23} = 0$: b is proportional to a and the factors are proportional, contrary to hypothesis. □

Theorem A.2 (uniform offsets)**.** *No such family exists with $b_i = -1$ for all i, nor with $b_i = +1$ for all i, for any $m \geq 2$.*

Proof. If all $b_i = -1$ then $D_{ij} = a_j - a_i$, the second product in (5) is $(-a_i)^{m-1}$, and the first becomes $\prod_{j \neq i}(2a_j - a_i) = a_i^{m-1}$. The a_i are distinct (non-proportionality). Take i with a_i maximal: for $j \neq i$, $0 < a_j < a_i$ gives $|2a_j - a_i| < a_i$, so the product has absolute value $< a_i^{m-1}$. The case $b_i = +1$ is the mirror image with $\prod_{j \neq i}(a_i - 2a_j)$. □

Theorem A.3 (generic position)**.** *Put $s_i = (b_i - 1)/a_i$ and $t_i = (b_i + 1)/a_i$. If the s_i are pairwise distinct and the t_i are pairwise distinct, no such family exists, for any $m \geq 2$.*

Proof. Let $G(\tau) = n(-\tau)$, of degree m with leading coefficient $c = (-1)^m \prod a_i$. The hypotheses say $G(s_i) = 1$ and $G(t_i) = -1$. With m distinct roots each, $G - 1 = c\prod(\tau - s_i)$ and $G + 1 = c\prod(\tau - t_i)$. Subtracting, $2 = c\big[\prod(\tau - t_i) - \prod(\tau - s_i)\big]$ identically in τ. The coefficient of τ^{m-1} on the right is $-c\sum(t_i - s_i) = -c\sum 2/a_i \neq 0$, while on the left it is 0. □

Corollary A.4. *Every CLC polynomial family has two indices $i \neq j$ with $p_j - 1 = \frac{a_j}{a_i}(p_i - 1)$ or $p_j + 1 = \frac{a_j}{a_i}(p_i + 1)$ as polynomial identities.*

The degenerate strata left open by Theorem A.3 were searched exhaustively using the fact that each factor in (5) divides a_i^{m-1}: no family exists with $m = 4$ and all $a_i \leq 500$, with $m = 5$ and $a_i \leq 150$, or with $m = 6$ and $a_i \leq 60$ (45 min, 1.8 h and 54 h respectively). Theorem A.2 was also confirmed by search for $m = 5, 7, 9, 11$ up to $a_i \leq 20000, 5000, 3000, 1500$.

Remark A.5. These theorems close families in which the divisibilities hold identically in k. They say nothing about families that are CLC only for k in some residue classes, nor about sporadic solutions. Theorem A.1 is the parametric shadow of Proposition 2.3.

A.2. **Two-sided subset products.** In the subset-product method one fixes a smooth modulus Λ, harvests primes p with $(p-1) \mid \Lambda$, and looks for a subset with product $\equiv 1 \pmod{\Lambda}$; the expected number of solutions is about $2^{(\text{pool size}) - \log_2 \varphi(\Lambda)}$, and the method succeeds because the pool grows faster than $\log_2 \varphi(\Lambda)$. For Williams numbers one needs simultaneously $(p+1) \mid \Lambda'$ and product $\equiv -1 \pmod{\Lambda'}$, and the two moduli interact: an odd prime ℓ cannot divide both some $p_i - 1$ and some $p_j + 1$ (else $n \equiv 1 \equiv -1 \pmod{\ell}$), so every small prime must be assigned to one side, and the 2-adic branch is fixed. Define the *margin* of a configuration as (number of usable primes) $- \log_2 \varphi(\text{lcm of all required } p \mp 1)$, so that the expected number of two-sided solutions is about 2^{margin}.

We measured the margin over three families of configurations: random smooth moduli (3,250 configurations, all with margin > -14 searched exhaustively by meet-in-the-middle, no solution); pools of all primes $p \leq X$ with both $p-1$ and $p+1$ y-smooth, with the side assignment optimised by hill-climbing (hundreds of restarts); and pools restricted to primes with few prime factors in $p \mp 1$. The best margins were -8.2 (tiny three-prime systems), $-29.0/-30.6/-31.0/-36.9$ for $y = 60/100/140/200$, and -36.9 for the sparse pools. The wall is flat or worsening in every direction we varied: each usable prime contributes one bit of subset freedom but about 2.5 bits of modulus, because a prime that is useful on one side is almost never useful on the other. This is a measurement, not a theorem. The side-assignment of the small primes was formalised by Zhang [26], who splits the prime factors of the modulus into two groups serving $p-1$ and $p+1$ respectively, conjectures that the number of Williams numbers built from a modulus with s prime factors grows only like $2^{2^{s^{1/2-o(1)}}}$ (against $2^{2^{s(1+o(1))}}$ for Carmichael numbers), and exhibits a set of 1,029 primes expected to contain at least 2^{24} Williams numbers, none of which has been found. Our margins are a small-scale measurement consistent with Zhang's conjecture and with the "intractable" judgement of [5].

Appendix B. A heuristic onset estimate

This appendix is heuristic. An Erdős–Pomerance-style count of k-factor products of primes with $p \mp 1$ both y-smooth, including the side-consistency factor $2(1-1/\ell)^k - (1-2/\ell)^k$ for each odd prime $\ell \leq y$ and a factor 2^{1-k} for the 2-adic branch, calibrated on the one-sided side against Pinch's counts $C(10^{15}) = 105{,}212$, $C(10^{18}) = 1{,}401{,}644$ and on the two-sided side against our measured counts of twin-smooth primes, places the first Williams number near 10^{208}–10^{490} with $k \approx 80$–110 and $y \approx 40$–90, and the first Lenstra–Pomerance counterexample to Agrawal's conjecture near 10^{2400}–10^{2900} with $k \approx 270$. The ranges reflect the sensitivity to the per-prime calibration; the qualitative conclusion—far beyond exhaustive search, reachable only by a construction that circumvents §A—is robust. The same machinery places the first Carmichael number near 10^3 (truth: 561).

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